PRACTICAL 5-B

Object Oriented Programming

AIM:

To model SavingsAccount, CurrentAccount and FixedDepositAccount through inheritance, abstract methods and run-time polymorphism.

CODE SCREENSHOT:

MiniBank.java

Part B > Practical 5-B > MiniBank.java > ...

```java
abstract class Account {
    protected String accountNumber, ownerName;
    protected long balance;

    Account(String n, String o, long b) {
        accountNumber = n;
        ownerName = o;
        balance = b;
    }

    abstract double interestRate();

    abstract boolean canWithdraw(long amount);

    public long getBalance() {
        return balance;
    }
}

class SavingsAccount extends Account {
    long minBalance;

    SavingsAccount(String n, String o, long b, long m) {
        super(n, o, b);
        minBalance = m;
    }

    double interestRate() {
        return 4.0;
    }

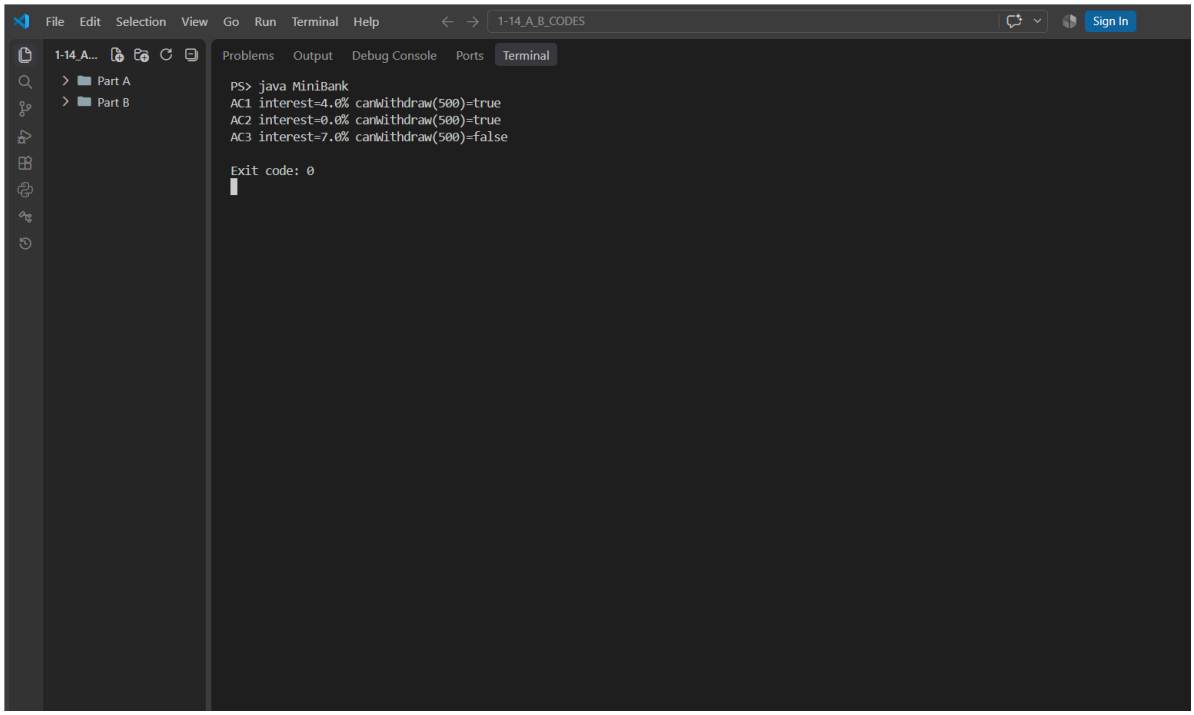
    boolean canWithdraw(long a) {
        return a > 0 && balance - a >= minBalance;
    }
}

class CurrentAccount extends Account {
    long overdraftLimit;

    CurrentAccount(String n, String o, long b, long l) {
        super(n, o, b);
        overdraftLimit = l;
    }

    double interestRate() {
        return 0.0;
    }
```

OUTPUT SCREENSHOT:



LEARNING OUTCOMES:

The student can design a class hierarchy and use polymorphism to write extensible code.